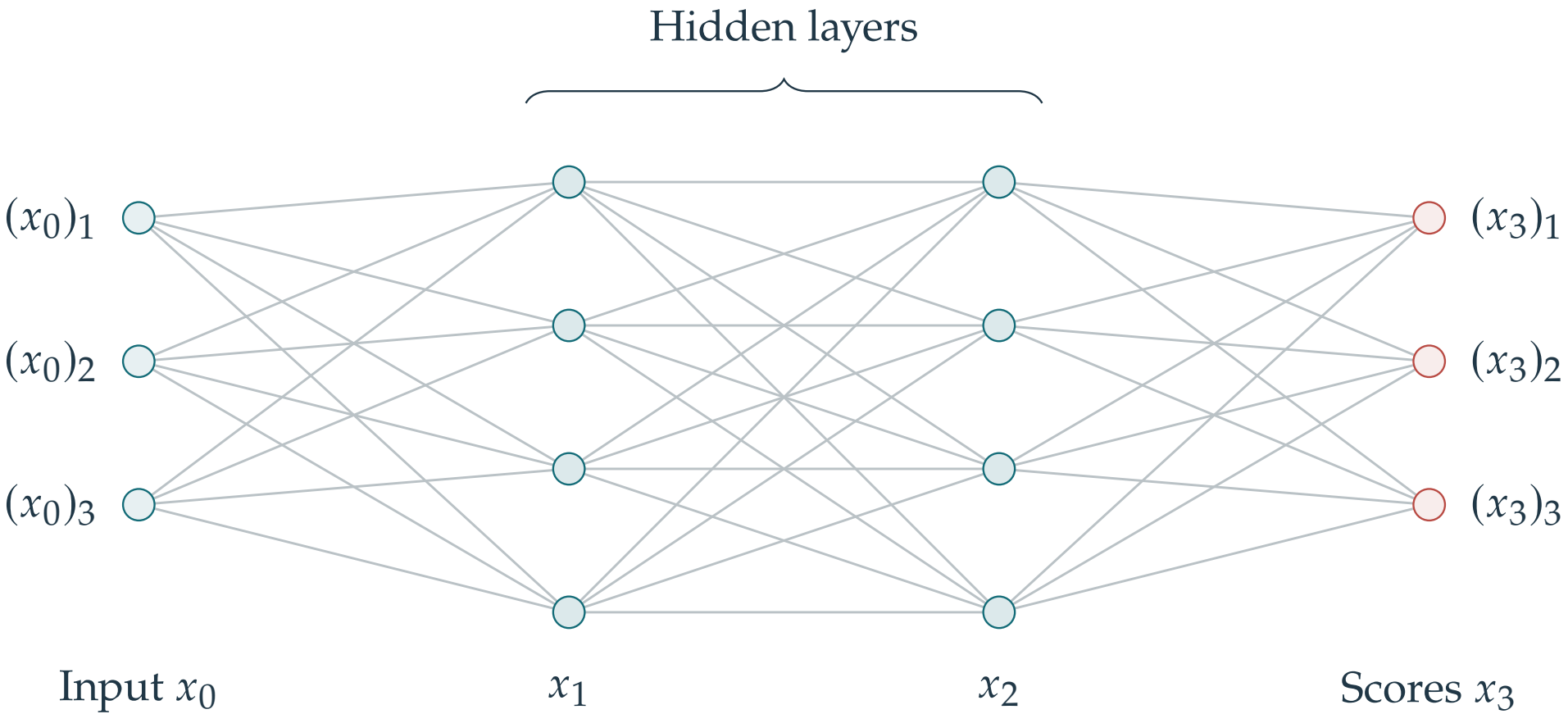

$$x_{\ell+1} = \rho_\ell(B_\ell x_\ell + b_\ell)$$